Symbolic Tumor-Growth ODE Candidate Ranking

Independent blinded evaluation; supplied training data only. Generated 2026-09-20.

Rank	Candidate ID	Main strength	Main concern	Reason for rank.
1	C9	Simple quadratic saturation with positive low-volume growth and declining high-volume growth.	Its fit is not the lowest-MSE among the accepted candidates.	It has a simple, smooth saturating form and the lowest training MSE among the low-complexity saturation candidates, with complexity 9.
2	C11	Clear positive-growth-to-saturation shape with a strong training fit.	The positive linear term makes growth initially increase with volume, which is less structurally simple than a monotone-declining saturation.	It retains a stable saturating shape while achieving a substantially lower training MSE (6.790523) than the simpler quadratic candidates.
3	C7	Simple quadratic saturation with a stable turnover from positive to negative growth.	Its training MSE is higher than C9 and C11.	Its robust saturating structure and complexity 9 make it a strong compromise, with MSE 7.314900.
4	C10	Monotone decreasing quadratic form with positive low-volume growth and saturation.	It is more complex than the simple quadratic candidates despite a higher MSE than C7/C9.	Its structure is robust and straightforward; MSE is 7.510162 with complexity 11.
5	C8	Simple quadratic saturation and low structural complexity.	Its training MSE is worse than C7 and C9.	Its robust form and complexity 9 keep it ahead of less straightforward higher-order candidates; MSE is 7.576442.
6	C3	Very simple linear positive-growth/stable-saturation model.	It cannot represent curvature in the growth-rate decline.	Its complexity 6 is attractive, but its MSE 7.603720 is above the better-fitting robust nonlinear candidates.
7	C1	Very simple linear positive-growth/stable-saturation model.	Its training MSE is slightly worse than C3.	It has the same robust linear structure and complexity as C3, so the supplied MSE separates them: 8.054396.
8	C12	Includes smooth curvature and a volume-dependent growth-rate transition.	The logarithm introduces a domain restriction and a non-monotone local growth-rate shape.	Its MSE 7.536199 is competitive, but the higher complexity 13 and log-domain consideration make it less straightforward than the simpler robust forms.
9	C14	Higher-order saturation with a finite stable positive equilibrium.	The cubic structure can introduce additional curvature and possible extra dynamical features.	Its low MSE (6.778854) is attractive, but criterion 1 places greater weight on the more complex higher-order shape.
10	C16	Higher-order form with very low training MSE and a finite stable saturation.	Its factored cubic structure permits more complicated volume-dependent behavior than the simpler saturating forms.	Its MSE 6.778712 is among the lowest, but complexity 16 and the added nonlinear structure reduce robustness/interpretability.
11	C18	Higher-order saturating polynomial with very low training MSE.	The higher-order polynomial permits more curvature and behavior away from the observed range.	Its MSE 6.762190 is the lowest in the table, but complexity 17 makes it less defensible under the ordered robustness/parsimony criteria.
12	C20	Higher-order polynomial with low training MSE and finite stable saturation.	Its fourth-order structure is more flexible and can create additional behavior outside the training region.	MSE 6.764555 is very low, but complexity 19 makes extrapolation less structurally constrained.
13	C13	Higher-order polynomial with a stable positive equilibrium.	The cubic term permits the high-volume behavior to change again, reducing extrapolative robustness.	Despite MSE 7.567556, its complexity 14 and more flexible shape place it below simpler candidates.
14	C17	Higher-order polynomial saturation with positive low-volume growth.	The cubic correction permits more complicated behavior away from the training range.	MSE 7.513984 is moderate, but complexity 17 offers little parsimony benefit.
15	C6	Linear stable-saturation model with the same simple structure as C3.	Its training MSE is worse than C3.	Its complexity 8 is higher than C3 while its MSE 7.603720 is identical to C3, so it is dominated on the supplied criteria.
16	C4	Linear stable-saturation model.	Its complexity is higher than C1 with essentially the same fit.	MSE 8.055493 and complexity 8 make it less defensible than the simpler C1.
17	C2	Simple linear stable-saturation model.	Its training MSE is relatively high.	Its complexity 6 is low, but MSE 8.168888 is the weakest among the complexity-6 linear candidates.
18	C5	Linear stable-saturation model.	It has the same basic structure as C2 but higher complexity.	MSE 8.168963 is essentially the same as C2 while complexity 8 is higher.
19	C15	Higher-order polynomial with a stable positive equilibrium.	The factored nonlinear form introduces substantial extra curvature and extrapolation flexibility.	MSE 7.562023 does not justify complexity 16 under the ordered criteria.
20	C19	Highest-complexity candidate with a very flexible polynomial structure.	Complexity 19 makes its extrapolation least constrained among the accepted set.	MSE 7.538104 is not sufficiently better than simpler candidates to offset its structural complexity.

Top choice: C9

C9 combines a simple quadratic saturation structure with positive growth at low volume and a stable turnover at higher volume, while also having the lowest training MSE among the low-complexity saturating candidates. Its complexity is substantially lower than the candidates with the very smallest MSE values, supporting a more constrained extrapolation without relying on resemblance to any known growth law.

The candidate set spans simple linear models, quadratic saturating forms, a logarithmic form, and increasingly flexible higher-order polynomials. The lowest training MSE values occur mainly in the higher-complexity models, but the prescribed ordering gives robustness and then parsimony priority, so fit improvements alone do not determine the ranking.

Trial information: Gaussian noise 15.0%; training interval [0,15]; forecast interval [15.5,25.5].